

Project Summary: Netflix Movie Database Analysis

Project Overview: This project involved an end-to-end data analysis of a movie database containing **9,827 entries**. The goal was to clean the raw data, perform exploratory data analysis (EDA), and visualize key trends such as genre distribution, popularity outliers, and movie ratings.

1. Data Inventory & Exploration

Initial inspection of the mymoviedb.csv file revealed a robust dataset with the following characteristics:

- **Total Rows:** 9,827
- **Total Columns:** 9 (Release_Date, Title, Overview, Popularity, Vote_Count, Vote_Average, Original_Language, Genre, Poster_Url)
- **Data Quality:** The dataset was initially very massy, contains many missing values.
- **Missing Values:** Inspection revealed **100 missing values** in the Vote_Average column, which were subsequently dropped to ensure statistical accuracy.

2. Data Cleaning & Feature Engineering

To prepare the data for visualization, several transformation steps were executed:

- **Temporal Conversion:** The Release_Date was converted to datetime64[us] format. The year was then extracted as an integer to facilitate chronological analysis.
- **Dimensionality Reduction:** Irrelevant columns for this specific analysis—Overview, Original_Language, and Poster_Url—were dropped to streamline the dataframe.
- **Genre Normalization:** The Genre column contained comma-separated strings (e.g., "Action, Adventure"). These were split into lists and "exploded," resulting in a new expanded dataset of **25,552 rows**, where every movie-genre pair occupies its own row.
- **Categorization:** Using a custom categorize_col() function, the Vote_Average was binned into four distinct categories based on quartiles: **not_popular**, **below_average**, **average**, and **popular**.

3. Statistical Insights

Through the df.describe() method, the following metrics were identified:

- **Average Rating:** The mean Vote_Average across the dataset is **6.44**.

- **Popularity Trends:** The mean popularity is **40.33**, but a massive standard deviation of **108.87** indicates significant outliers.
- **Max Popularity:** The most popular movie in the dataset is "**Spider-Man: No Way Home**" (2021), with a popularity score of **5083.95**.
- **Vote Distribution:** The categorizing process resulted in a balanced distribution across the four rating tiers, with "not_popular" holding 2,467 entries and "popular" holding 2,450.

4. Key Visualizations & Findings

A. Genre Dominance

Analysis of the 19 unique genres revealed that **Drama** is the most frequent genre in the database, appearing **3,715 times**. Other highly frequent genres include Comedy and Action.

B. Historical Production Trends

A histogram of the Release_Date shows that movie production has seen an exponential increase over the decades, with the vast majority of the 9,827 movies being released between **2010 and 2022**.

5. Conclusion

The analysis successfully transformed a raw CSV into a clean, categorical dataset. The findings highlight a modern-leaning database where **Drama** is the king of content, and high-budget blockbusters like **Spider-Man: No Way Home** represent extreme statistical outliers in terms of global popularity.